



MTN Liquidation API Doc V 1.0

CONFIDENTIALITY DISCLAIMER

DEVELOPER API GUIDE DOCUMENT

PROJECT: Equity IMT Outbound On Premise

API Document

Date: 12.04.2022

THIS DOCUMENT AND THE INFORMATION IN IT ARE PROVIDED IN CONFIDENCE, FOR THE SOLE PURPOSE OF EQUITY BANK LIMITED AND MAY NOT BE DISCLOSED TO ANY THIRD PARTY OR USED FOR ANY OTHER PURPOSE WITHOUT THE EXPRESS WRITTEN PERMISSION OF EQUITY BANK LIMITED.

Any form of reproduction, dissemination, copying, disclosure, modification, distribution and or publication of this material is strictly prohibited. The contents of this document have not been independently verified and they do not purport to be comprehensive, or to contain all the information that a prospective investor may need. No representation, warranty or undertaking, expressed or implied is or will be made or given and no responsibility or liability is or will be accepted by Equity Bank Limited or by any of its directors, employees or advisors in relation to the accuracy or completeness of this document or any other written or oral information made available in connection with the Company.

Revision History

Date	Version	Author	Comments
29-08-2022	v1.0	Kennedy Gatimu	Initial document creation

Table Of Contents	
1.0 NAME CHECK SERVICE	5
1.0.1 Name Check Request Fields And Description	5
1.0.2 Name Check Response Fields And Description	6
1.0.3 Name Check Status Possible Values	6
1.0.4 Name Check Sample Request	6
1.0.5 Name Check Sample Response	6
2.0 DEBIT/ DEPOSIT SERVICE	7
2.0.1 Debit Request Fields And Description	7
2.0.2 Debit Response Fields And Description	8
2.0.3 Debit Status Possible Values	8
2.0.4 Sample Debit Request	9
3.0 TRANSACTION STATUS CHECK SERVICE	10
3.0.1 Transaction Status Check Request Fields And Description	10
3.0.2 Transaction Status Check Response Fields And Description	11
3.0.3 Sample Transaction Status Check Request	11
3.0.4 Sample Transaction Status Check Response	11
3.0.5 Transaction Status Check Possible Values	12
4.0 DIGITAL SIGNATURE GUIDE	13
4.0.1 Generating The Signature	13
4.0.2 Sign The Clear Text	13
4.0.3 Sample Java Code	14

1.0 NAME CHECK SERVICE

Method	POST
URL	https://10.11.6.111:8080/api/v1/namecheck
Headers	Content-Type:application/xml

1.0.1 Name Check Request Fields And Description

NO	FIELD	DESCRIPTION	MANDATORY	TYPE
1	mobileNumber	Destination mobile. Allowed Mobile Number Length Is 12 Characters Excluding Plus (+) Suffix. Example - 256779000000	Yes	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	Yes	String
3	msgId	Reference number	Yes	String
4	telcom	Name of the destination Telco Example: MTN-UG	Yes	String
5	channel	Channel code Example: MFS	Yes	String
6	signature	Digital signature. See Digital signature section for more information.	Yes	String
7	nationalID	Id of the receiver. If not, supply 000000	Yes	String

1.0.2 Name Check Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	msgId	Reference number	String
2	channel	Channel code	String
3	mobileNumber	Destination mobile.	String
4	responseCode	Response code	String
5	customerNames	Unique identifier for the transaction	String

1.0.3 Name Check Status Possible Values

RESPONSE CODE	DESCRIPTION
00	Name check request successful
01	Name check request failed

1.0.4 Name Check Sample Request

```
<?xml version="1.0" encoding="UTF-8"?>
<mobileNumberInquireReq>
  <mobileNumber>256779999703</mobileNumber>
  <timestamp>26-Aug-2022 12:55:32</timestamp>
  <msgId>415976</msgId>
  <telcom>MTN-UG</telcom>
  <channel>MFS</channel>
  <signature>QmxK19mYaXgLx4+dWTxetKLdkFSD6SpF5gaPER41+Q==</signature>
  <nationalID>000000</nationalID>
</mobileNumberInquireReq>
```

1.0.5 Name Check Sample Response

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<MobileNumberInquireRes>
  <msgId>415976</msgId>
  <channel>MFS</channel>
  <mobileNumber>256779999703</mobileNumber>
  <responseCode>00</responseCode>
  <customerNames>Martha A</customerNames>
</MobileNumberInquireRes>
```

2.0 DEBIT/ DEPOSIT SERVICE

Method	POST
URL	https://10.11.6.111:8080/api/v1/deposit
Headers	Content-Type:application/xml

2.0.1 Debit Request Fields And Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	msgld	Reference number	Y	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	Y	String
3	rrn	Retrieval reference number (Can be same as msgld)	Y	String
4	amount	Transaction amount rounded off to 2 decimal places. Example: 100.00	Y	String
5	currencyCode	Currency	Y	String
6	tranType	Type of Transaction. Example: WalletToAccount	Y	String
7	receiverMsisdn	Destination mobile. Allowed Mobile Number Length Is 12 Characters Excluding Plus (+) Suffix. Example - 256779000000	Y	String
8	receiverName	Name of the receiver. Usually from Name Check service	Y	String
9	senderName	Sender name	Y	String
10	senderMsisdn	Sender mobile. Allowed Mobile Number Length Is 12 Characters Excluding Plus (+) Suffix. Example - 256779000000	Y	String
11	callbackURL	Sending channel callback url. Used to send final transaction status to the calling channel.	Y	String
12	signature	Digital signature. See Digital signature section for more information.	Y	String

2.0.2 Debit Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	msgId	Reference number	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	String
3	rrn	Retrieval reference number	String
4	status	Transaction status	String
5	responseDesc	Transaction status description	String
7	transactionRef	Unique identifier for the transaction	String

2.0.3 Debit Status Possible Values

STATUS(status)	DESCRIPTION
ACK	Debit request was accepted by MTN awaiting the final status
FAILURE	Debit request failed

2.0.4 Sample Debit Request

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<EazzyDepositReq>
  <msgId>264375443725</msgId>
  <timestamp>2022-08-26T15:39:22.518</timestamp>
  <rrn>264375443725</rrn>
  <amount>2000.00</amount>
  <currencyCode>UGX</currencyCode>
  <tranType>WalletToAccount</tranType>
  <receiverMsisdn>256702837908</receiverMsisdn>
  <receiverName>NABAKOOZA DEBORAH</receiverName>
  <senderName>Test Customer</senderName>
  <senderMsisdn>256788566300</senderMsisdn>
  <callbackURL>http://10.11.6.59:7007/mfmbisug/mbintf/mtnCallbackService.jsp</callbackURL>

  <signature>QJmj6JGeznXt06YrYtpOFp+4jwn3mwUgal3ySLkoVf3PcAOehVyrC9v8qqjrj8L8x9i1zxW0v
ya+gpa+ltHVB58wjmTJdKKB9y+qDi1DFfaBlpqYthO1Oksu0QNkWwaGWy3hc5UgXS53Nh8yLNKsrE
C5KoCZ2wld6coOvoDIMLts5v2qEJ0HwZ7LuU4aKi2GwvV6Xbz+MXwLzv4f/gV92oBhDjsoa51kIN5kw
cz0nOahJLaIkchFD3gC7JLdrS8Yfhndy66JUfxmEBU5ZQnToUDxfZpSO8pwCxlTPoSWCGpVjpncE
acZMemSXkYVzXkqrLGzqSHoAV/NqdlhfnUVfg==</signature>
</EazzyDepositReq>
```

2.0.5 Sample Debit Response

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<EazzyDepositResp>
  <msgId>264375443725</msgId>
  <timestamp>2022-08-26T15:39:23.257</timestamp>
  <rrn>264375443725</rrn>
  <status>ACK</status>
  <responseDesc>Request Received By Telco. Waiting Final Status. Initial Telco Status:
    PENDING</responseDesc>
  <transactionRef>4580503</transactionRef>
</EazzyDepositResp>
```

3.0 TRANSACTION STATUS CHECK SERVICE

Method	POST
URL	https://10.11.6.111:8080/api/v1/status-check
Headers	Content-Type:application/xml

3.0.1 Transaction Status Check Request Fields And Description

NO	FIELD	DESCRIPTION	MANDATORY	TYPE
1	rrn	Retrieval reference number (Can be same as msgld)	Yes	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	Yes	String
3	msgld	Reference number	Yes	String
4	telcom	Name of the destination Telco Example: MTN-UG	Yes	String
5	channel	Channel code Example: MFS	Yes	String
6	signature	Digital signature. See Digital signature section for more information.	Yes	String

3.0.2 Transaction Status Check Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	msgld	Reference number	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	String
3	rrn	Retrieval reference number	String
4	status	Transaction status	String
5	responseDesc	Transaction status description	String
7	transactionRef	Unique identifier for the transaction	String

3.0.3 Sample Transaction Status Check Request

```
<?xml version="1.0" encoding="UTF-8"?>
<transactionStatusInquireReq>
  <rrn>179003244933</rrn>
  <timestamp>26-Aug-2022 12:55:32</timestamp>
  <msgld>179003244933</msgld>
  <telcom>MTN-UG</telcom>
  <channel>MFS</channel>
  <signature
>QmxK1FFcHwCwylhplvCi7V4SLPTMO9mYaXxgLx4+dWTxetKLdkFSD6SpF5gaPER41+
</signature>
</transactionStatusInquireReq>
```

3.0.4 Sample Transaction Status Check Response

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<EazzyDepositResp>
  <msgld>179003244933</msgld>
  <timestamp>2022-08-29T11:55:50.085</timestamp>
  <rrn>179003244933</rrn>
  <status>PENDING</status>
  <responseDesc>Request Timeout Or Service Not Available At The Moment</responseDesc>
  <transactionRef>4580512</transactionRef>
</EazzyDepositResp>
```

3.0.5 Transaction Status Check Possible Values

STATUS(status)	DESCRIPTION
SUCCESS	Debit request was accepted by MTN awaiting the final status
FAILURE	Debit request failed
PENDING	Transaction final status is still not known
EXPIRED	Transaction is expired

4.0 DIGITAL SIGNATURE GUIDE

For the purpose of ensuring data integrity and security, every channel will be required to generate a digital certificate and share the same with the maintainer of this API. The certificate will be used to verify the signature contained in each payload.

4.0.1 Generating The Signature

Each request should have a digital signature generated as described below:

1. Generate the clear text:

Concatenate the values in each request field shown below to generate the clear text. The clear text value will be hashed.

a. Clear Text For Name Check

```
clearText = mobileNumber  
          + msgId  
          + telcom  
          + channel
```

b. Clear Text For Debit

```
clearText = msgId  
          + rrn  
          + timestamp  
          + amount  
          + currencyCode  
          + tranType  
          + receiverMsisdn  
          + senderMsisdn
```

Note: The amount is a big-decimal value rounded off to a scale of 2, Ceiling

b. Clear Text For Transaction Check Status

```
clearText = rrn  
          + msgId  
          + telcom  
          + channel
```

4.0.2 Sign The Clear Text

a. Generate RSA private key

b. Using the private key and SHA-256 hashing algorithm, sign the clear text generated above.

In summary, the signature scheme should be SHA-256 with RSA

4.0.3 Sample Java Code

The figure below shows an example of how the signature can be generated using Java although the same can be achieved using any programming language

Example of how to generate a digital signature using Java

```
import java.security.PrivateKey;
import java.security.Signature;
import java.util.Base64;
public class SignMsg {
    private static final String UTF_8 = "UTF-8";
    private static final String ALGORITHM = "SHA256WithRSA";
    public String signMessage(String clearText , PrivateKey privatekey) {
        try {
            byte[] messageBytes = clearText .getBytes(UTF_8);
            Signature signatureInstance = Signature.getInstance(ALGORITHM);
            signatureInstance.initSign(privatekey);
            signatureInstance.update(messageBytes);
            byte[] digitalSignature = signatureInstance.sign();
            String signature = Base64.getEncoder().encodeToString(digitalSignature);
            return signature;
        } catch (Exception ex) {
            ex.printStackTrace();
            return null;
        }
    }
}
```